Code No: 111AE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, June - 2022
ENGINEERING CHEMISTRY
(Common to CE, EEE, ME, ECE, CSE, IT, MMT, AE, MIE, AGE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) What is a reference electrode? Discuss the construction of calomel electrode and calculate the pH of unknown solution using calomel electrode.
b) Discuss the electrochemical theory of corrosion and explain the mechanism of rust formation in acidic and neutral medium. [8+7]

2.a) Explain the discharging and charging process of a lead acid storage battery with reactions.
b) Discuss the principle involved in cathodic protection method and describe the two types of cathodic protection methods. [7+8]

3.a) What are conducting polymers ? Discuss the classification, mechanism and applications of conducting Polymers.
b) Explain the synthesis, properties and applications of
(i) PVC (ii) Bakelite (iii) BUNA – S rubber [6+9]

4.a) Explain the setting and hardening steps of cement with relevant chemical reactions involved.
b) What are Lubricants? How are they classified? Give examples. [8+7]

5.a) Describe the ion exchange method used for softening of water with a neat diagram.
b) Discuss about the following:
i) Scale and Sludge formation
ii) Caustic embrittlement [9+6]

6.a) What is the principle of EDTA method? Describe the estimation of Hardness of water by EDTA method.
b) Discuss Break point chlorination. [9+6]

7.a) Discuss proximate method of analysis of coal sample and give its significance.
b) What is cracking? Explain fixed bed catalytic cracking process and its advantages. [8+7]

8.a) Explain one component system phase diagram using water system.
b) What are colloids? How are they classified? Mention its properties and applications. [8+7]

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